

# Chapter 20 - Hibernate & JPA Interview Questions

## Hibernate Fundamentals

1. What is Hibernate?
2. What is ORM?
3. Advantages of Hibernate?
4. Hibernate vs JDBC?
5. Hibernate Architecture?

## Session & Transactions

6. Difference between Session and SessionFactory?
7. Why is Session not thread-safe?
8. openSession() vs getCurrentSession()?
9. What is Transaction?
10. flush() vs commit()?

## Entity Lifecycle

11. What are Transient, Persistent, Detached and Removed states?
12. update() vs merge()?
13. What is Persistence Context?
14. What is Dirty Checking?

## Fetching & Caching

15. get() vs load()?
16. get() vs getReference()?
17. What is a proxy object?
18. Explain Level 1 Cache.
19. Explain Level 2 Cache.
20. Does Level 1 Cache cache null?
21. Query Cache vs Level 2 Cache?

## Mappings & Annotations

22. @Entity vs @Table?
23. @Transient vs transient?
24. @Id and @GeneratedValue strategies?
25. mappedBy purpose?

- 26. Owning side?
- 27. OneToMany vs ManyToOne?

## **Fetch & Cascade**

- 28. LAZY vs EAGER?
- 29. Default fetch types?
- 30. CascadeType.ALL?
- 31. LazyInitializationException?
- 32. N+1 Query Problem?

## **Queries**

- 33. HQL vs SQL?
- 34. Criteria API vs HQL?
- 35. Native SQL?
- 36. Fetch Join?
- 37. Pagination?

## **JPA**

- 38. What is JPA?
- 39. JPA vs Hibernate?
- 40. EntityManager vs Session?
- 41. EntityManagerFactory vs SessionFactory?

## **Spring Integration**

- 42. Why @Transactional on Service?
- 43. Controller vs Service vs Repository?
- 44. How Spring integrates with Hibernate?

## **Scenario Questions**

- 45. How would you optimize slow Hibernate queries?
- 46. How do you handle concurrent updates?
- 47. When do you use Level 2 Cache?
- 48. How do you process one million records?
- 49. Explain a production issue you fixed using Hibernate.

## **Interview Preparation Tips**

Understand concepts instead of memorizing APIs. Be able to explain entity lifecycle, persistence context, dirty checking, caching, transactions, fetch strategies, and common production issues with

practical examples.